

PROCES VERBAL DE SOUTENANCE DE DOCTORAT

à déposer dans les 3 jours ouvrés après la soutenance au service de scolarité de l'établissement de préparation du doctorat daté et signé

Monsieur Pablo DONATO

ECOLE DOCTORALE : Ecole Doctorale de l'Institut Polytechnique de Paris

Titre de la thèse : Inférence Profonde pour la Démonstration Graphique de Théorèmes

Spécialité de doctorat : Informatique

Date de soutenance : 29 mai 2024 Heure : 14h00 Lieu : Salle Gilles Kahn LIX - UMR 7161 1 rue Honoré d'Estienne d'Orves

Bâtiment Alan Turing Campus de l'École Polytechnique

Décision du Jury

- ☒ Admission
☐ Ajournement

A l'issue de la soutenance Monsieur DONATO Pablo a prêté serment : ☐ Oui ☐ Non

Avis du Jury sur la reproduction de la thèse :

- ☒ Thèse dont le dépôt légal peut être finalisé après des modifications mineures
Des modifications mineures seront toujours nécessaires au moins pour mentionner sur la page de couverture le nom du président du jury en vue du 2nd dépôt légal de la thèse.
- ☐ Corrections majeures demandées par le jury dans un délai de 3 mois
Si des corrections majeures sont demandées, identification du **membre du Jury désigné par le président** pour vérifier les corrections :
Nom, prénom, titre et fonction dans le jury :

Dans ce cas, le président du jury complète, date et signe le formulaire de vérification des corrections majeures apportées à la thèse et le remet au membre du jury chargé de la vérification.

Président du jury : _____

Civilité, Nom, Prénom	Titre	Fonction dans le Jury	Président	Signature
M. Benjamin WERNER	Directeur de recherche	Directeur de thèse		Le directeur de thèse, ainsi que toute autre personne ayant participé à la direction de la thèse, ne prend pas part à la décision*
M. Nicolas MAGAUD	Professeur des universités	Rapporteur	☐ Président	N. Magaud
M. Anupam DAS	Associate Professor	Rapporteur		Anupam
M. Paul-André MELLIÈS	Directeur de recherche	Examineur	☐ Président	
M. Hugo HERBELIN	Directeur de recherche	Examineur	☒ Président	H. Herbelin
M. Pierre-Yves STRUB	Research Scientist	Co-encadrant de thèse		Le directeur de thèse, ainsi que toute autre personne ayant participé à la direction de la thèse, ne prend pas part à la décision*
M. Serge AUTEXIER	Directeur de recherche	Examineur	☐ Président	S. Autexier
M. Laurent THÉRY	Chargé de recherche	Examineur		L. Théry

* Article 18 de l'arrêté du 25 Mai 2016, modifié par l'arrêté du 26 août 2022

RAPPORT DE SOUTENANCE DE DOCTORAT

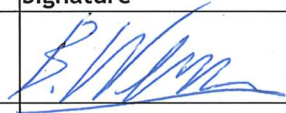
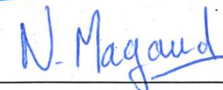
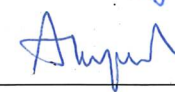
à déposer si possible trois jours ouvrés après la soutenance et au plus tard un mois après la soutenance au service de scolarité de l'établissement de préparation de la thèse, daté et signé

Monsieur Pablo DONATO

ECOLE DOCTORALE : Ecole Doctorale de l'Institut Polytechnique de Paris

Pablo Donato a présenté ses travaux combinant conception de langages de preuve graphiques et développement d'une interface gestuelle élégante et innovante pour le raisonnement logique (Actema). Cette approche exploite la possibilité d'inférence sous les connecteurs (inférence portée) et est susceptible de déboucher vers une nouvelle génération d'interfaces pour les assistants à la preuve, qu'il a déjà implémenté en Coq.

L'exposé, pédagogique, synthétique, et d'un grande maturité, a mis en valeur un travail original, de très grande ampleur, convaincant et maîtrisé, exploitant un panorama remarquablement large de la littérature existante, aux implications à la fois théoriques et pratiques, ce dernier aspect ayant été illustré par de nombreux exemples d'utilisation d'Actema. Le jury a apprécié la grande qualité des réponses aux très nombreuses questions, qu'elles soient techniques, ou portant sur les perspectives de son travail. Les membres du jury, unanimes, sont convaincus que Pablo Donato a toutes les qualités pour poursuivre une brillante carrière de chercheur.

Civilité, Nom et Prénom	Titre	Fonction dans le Jury	Président	Signature
M. Benjamin WERNER	Directeur de recherche	Directeur de thèse		
M. Nicolas MAGAUD	Professeur des universités	Rapporteur	☐ Président	
M. Anupam DAS	Associate Professor	Rapporteur		
M. Paul-André MELLIÈS	Directeur de recherche	Examineur	☐ Président	
M. Hugo HERBELIN	Directeur de recherche	Examineur	☒ Président	
M. Pierre-Yves STRUB	Research Scientist	Co-encadrant de thèse		
M. Serge AUTEXIER	Directeur de recherche	Examineur	☐ Président	
M. Laurent THÉRY	Chargé de recherche	Examineur		

Report on Pablo Donato’s PhD thesis: *Deep Inference for Graphical Theorem Proving*

Reviewer: Anupam Das

May 2024

Context and significance

Proof theory is the study of mathematical proofs as formal objects in their own right. *Structural* proof theory is an umbrella term for proof theoretic activity where notions of inference step, normal forms, and equality of proofs have primary consideration. It is not a single area in its own right, but rather encompasses quite different technical developments seeking to understand and model proofs under the aforementioned considerations.

The foundational results of structural proof theory were Gentzen’s famous normalisation theorems in the 1930s. His *natural deduction* systems underlie the famous *Curry-Howard* correspondence, identifying formal proofs with functional programs, itself the basis of modern proof assistants such as Coq.

Over the course of almost 100 years since Gentzen’s work, countless alternative proof formalisms have been proposed, leading to now a rich landscape of systems within the discipline of structural proof theory. Two recent themes of notable interest are:

1. graphical formalisms (e.g. Pierce’s existential graphs, Andrews’ matings, Miller’s expansion trees, Girard’s proof nets, Guglielmi’s atomic flows, Hughes’ combinatorial proofs,...); and,
2. ‘deep inference’ formalisms (e.g. hyper/nested sequents, calculus of structures, open deduction,...).

Both of these themes have significantly advanced the capacity of structural proof theory, in particular providing alternative intuitions for formal proof building and, indeed, exhibiting *more* proofs than Gentzen’s systems, in a formal sense. However, importing these ideas into the activity of *theorem proving* has remained underdeveloped, despite the promise they show to this end. Over the course of the last 20 years, only glimpses of such applications have appeared. This PhD thesis is a welcome shot in the arm with respect to these pursuits, amending an unfortunate oversight of the literature.

Summary of contributions

The guiding principle of this thesis is Proof-by-Action (PbA). This is inspired by the concept of *direct manipulation* in user interfaces dating back to the '80s, which has seen little development in the world of theorem provers. Nonetheless PbA seeks to bring together previous approaches such as Proof-by-Pointing and Proof-by-Linking. In this thesis there is particular focus on *click* actions and *drag-and-drop* actions. A prototype implementation *Actema* is given, a user interface incorporating click and DnD actions, and serves as a case study of recurring attention in the thesis.

Proof-by-Linking is based on the idea of subformula linking from *linear logic*. This thesis develops these ideas proof theoretically, in particular developing appropriate *deep inference* systems of intuitionistic and classical predicate logic. These systems appropriately model actions in Actema, thus establishing the soundness of that interface. The extension to classical logic incorporates modern ideas from the semantics of classical logic (as a computational system), where disjunction is modelled as a sort of parallel interaction.

The thesis explores several use cases of Actema and techniques in the PbA paradigm. Notably it also develops the *Coq-Actema* system, recasting Actema as a front-end to the Coq proof assistant. This works by interpreting the actions of Actema as novel tactics in Coq by way of a plugin. While this system still has some way to go, its development is a welcome contribution and has the potential for significant future improvements.

In the second part, this thesis develops graphical calculi inspired by various forms of deep inference, notably nested sequents and Pierce's existential calculi. While motivated by the PbA paradigm, this part seems more focussed on theory building. It introduces two formalisms: the *bubble calculus* and the *flower calculus*. The development of both formalisms is governed by a discipline of 'iconic manipulation', favouring graphical and 'metaphorical' notations that are suggestive of the objects they represent.

The bubble calculus is inspired by Brünnler's nested sequents. A calculus BJ is given for intuitionistic logic, where the metaphor of bubbles is suggestive of context management. Notably, and reassuringly, the bubble calculus readily extends to desirable fragments and extensions, in particular classical logic and bi-intuitionistic logic. A general soundness result is given for these systems by interpretation within their algebraic formulations. Completeness for the bi-intuitionistic calculus is established by simulation of an existing system, whence completeness for the other calculi is recovered.

The flower calculus is an alternative calculus for intuitionistic logic, whose notations/concepts are given according to a horticultural taxonomy. It follows a rewriting style discipline inspired by Pierce's existential graphs, as well as deep inference in the main. This is preceded by a modern and comprehensive account of the theory of existential graphs themselves. Soundness and completeness of Kripke relational structures is established, the latter in particular employing semantic/model theoretic techniques. This actually simultaneously establishes

analytic completeness, adapting an earlier argument of Hermant for cut-free sequent calculi. Some ideas for proof search algorithms are presented, and more notably an implementation of the flower calculus in the *Flower Prover* is given, exploiting the the metalogical adequacy of the flower calculus.

Evaluation

This PhD thesis has been a pleasure to read. Pablo’s writing exhibits a level of maturity that even senior researchers rarely accomplish. In particular the careful structuring of the content, the deep literature survey, and generally helpful presentation (e.g. margin notes, use of colour/shading, internal hyperlinking), comprise academic writing of outstanding quality.

The merits of the thesis’ presentation do not come at the cost of its content, with the manuscript standing at almost 300 pages! This is not simply a matter of quantity: Pablo exhibits deep dives into several quite different technical directions, and moreover develops new theory and tools at their interfaces. The breadth of material in this manuscript could easily cover two PhDs.

I would particularly highlight that Pablo’s innovation in recasting ideas from graphical proof theory and deep inference as tactics and interfaces for proof assistants is very welcome. This is not a matter of simply plugging things together, rather requiring innovation all the way from foundations to implementation.

Further comments

Some of the more philosophical motivation underlying ‘iconic manipulation’ was somewhat lost on me. In particular, I am not wholly convinced that this discipline facilitates comprehension of proofs. For instance I could not easily appreciate the commitment to horticultural taxonomy for the flower calculus. Of course I look forward to debating this matter with Pablo soon.

On the proof theoretic side, it would be particularly interesting in future work to develop the Curry-Howard aspects of the calculi proposed, in particular reductive cut-elimination and term annotations. Pablo has already outlined some of these ideas for the flower calculus. Such considerations are generally underdeveloped for deep inference and graphical calculi as a whole.

Recommendation and conflict declaration

Pablo Donato should be awarded a PhD on the basis of the thesis submitted. I confirm that I, the undersigned, do not have any conflict of interest to declare in relation with this thesis.



Dr Anupam Das



Review of Pablo Donato's doctoral dissertation *Deep Inference for Graphical Theorem Proving*

Nicolas Magaud (magaud@unistra.fr)

Professeur des Universités

Lab. ICube UMR 7357 CNRS Université de Strasbourg, France

Writing formal proofs in Interactive Theorem Provers (ITPs) such as Coq remains very technical and requires a high level of expertise. Indeed, the users have to master not only the scientific concepts (mathematics, computer science, physics, etc.) on which their proof relies, but also all the specific and sometimes intricate features of the logic implemented in their favorite ITP. Interfaces to develop proofs are usually rather basic and only provide minimal textual assistance for ITPs users.

Making proofs easier to achieve on a computer, especially through a comfortable and intuitive user interface, is useful both for beginners and for expert users. For beginners, it provides an enjoyable first experience and makes the proving process closer to pen-and-paper proofs. For expert users, it could significantly improve their productivity. Bringing (graphical) user interfaces for ITPs at the same level as in other fields of computer science is therefore unavoidable and is a key feature to speed up the adoption of proof assistants and make them a realistic alternative to pen-and-paper proofs in mathematics or computer science.

In his thesis, Pablo Donato deals with the issue of improving the way people interact with a proof assistant. He studies how to rely on a graphical interface rather than a textual one, introducing *graphical theorem proving*, especially through deep inference which allows to handle subformulas at an arbitrary depth in the goals.

The thesis is divided into 2 main parts: the first one deals with symbolic manipulations and explores the logical behavior of graphical actions whereas the second one deals with iconic manipulations and presents some original logical frameworks which aim at capturing the gestural approach to proof building from a proof theoretical perspective.

The first part of the thesis presents a prototype implementation (called *Actema*) of a graphical interface for theorem proving, which could replace the usual text-based approach used for about 30 years. For practical reasons and to be able to apply his ideas to several existing proof systems, Pablo focuses on first order logic. His work is inspired by the seminal work known as *proof by pointing* introduced by Bertot, Kahn and Théry and generalizes more recent work, namely *proof by linking*, initiated by Chaudhuri. Overall, Pablo names the approach that he follows *proof by action*. Indeed, he aims at replacing existing textual proof languages with gestural actions, which could be performed directly in a graphical user interface.

In Chap. 2, Pablo studies how both *click* and *drag-and-drop* (DnD) actions on some formulas of a sequent behave. It requires translating these actions into actual operations of the underlying logic, namely showing how to deal with logical connectives, universal and existential quantifications and equality. Pablo shows that deep inference allows to carry out more involved proof steps such as drag-and-drop actions through connectives, thus leading to elaborating more complex tactics in the background. In Chap. 3, the logical semantics of drag-and-drop actions are defined in the setting of deep inference proof theory. Following the subformula linking principles of Chaudhuri, he proposes to add a validity condition on linkages to avoid unproductive DnD actions. Under this condition, DnD actions are only permitted when the subformulas at stake are unifiable. In Chap. 4, Pablo shows, using three simple but carefully-chosen examples with an educational flavor, how actions can be used to carry out proofs. These three examples are representative of what can be achieved through the proof by action paradigm and deal with first-order logic, sets, relations and functions as well as Peano arithmetic. For each statement, the graphical proof by action comes together with an *equivalent* Coq proof script. It is worth noting that proof by action allows for both backward and forward reasoning, whereas common proof assistants usually strongly favor backward reasoning, making it hard to understand for beginners with little or no background in logic. In Chap. 5, Pablo advocates that an unusual feature in interactive theorem proving, namely multi-conclusion sequents, can be easily captured by the proof by action paradigm. He argues that direct manipulation through gestural actions makes it easy to deal with multiple conclusions, and introduces a new operator to model reasoning in classical logic that involves the interaction of several conclusions. Finally, in Chap. 6, Pablo explains how the proof-by-action principles can be implemented for first order logic as a web application (*Actema*). He also presents a prototype extension (*coq-actema*) to the Coq proof assistant, based on a plugin, which allows for direct manipulations of the proof state of Coq through gestural actions.

The second part of the thesis deals with iconic manipulations and explores new deep inference proof systems to give more structure to the notion of

goal. It consists in 4 chapters, one for each proposed system. These systems allow for both textual and graphical representations of the goals. The textual representation is well-suited for the backend of an ITP whereas the graphical representation, described in a so-called *metaphorical notation* (bubbles or flowers) is well suited for direct manipulation and thus for the frontend of an ITP (its graphical user interface).

Chap. 7 and 8 introduce the Asymmetric and Symmetric Bubble Calculi, which are extensions of nested sequents and can be imagined as rewriting systems with a graphical interpretation. In Chap. 8, the concept (called the *metaphor* in the thesis) of bubbles and an associated calculus for intuitionistic logic (BJ) are introduced. They capture the separation and sharing of contexts between different subgoals. In Chap. 8, the calculus BJ is extended into a more general and symmetric calculus (B) for classical logic. A variant of this calculus, in which rules are all invertible (which is useful to ensure goals remain provable after each gestural action) is also proposed. In Chap. 9, existentials graphs, introduced by Pierce in the late nineteenth century/early twentieth century, are presented from an historical perspective and in a self-contained way, paving the way to Flower Calculus introduced in the next chapter. Existential graphs allow to achieve *full iconicity*, i.e. that every logical construction has an associated icon. Therefore, in this setting, connectives and qualifiers of symbolic formulas become useless.

The last chapter of this thesis (Chap. 10) introduces the principles of the Flower Calculus, a new proof system for intuitionistic predicate logic whose syntactic objects are called flowers. The Flower Calculus is the last of a sequence of calculi, inspired by Pierce's existentials graphs. Meta-theoretical properties of this new Calculus are investigated. Following the flowers metaphor, the system is splitted into a *natural* fragment (where every rule is analytic and invertible) and a *cultural* fragment (where every rule is non-invertible). Pablo shows the completeness of the natural fragment and uses this property to show that the cultural fragment is admissible. These meta-theoretical results are then put in practice to design the Flower Prover, a new Graphical User Interface which aims at unifying the notions of goal and theory. Goals are flowers, which are manipulated with the natural rules of the calculus whereas theories are the same flowers manipulated with the cultural rules. Finally, the author draws some perspectives, especially investigating how to ensure the proof search procedure in this framework can be shown complete and terminating.

Overall this thesis presents not only some remarkable contributions to proof theory with a gestural-actions (i.e. graphical user interface) perspective but also some significant practical ones. Pablo proposes new original approaches to ease the interaction between humans and computers to carry out formal proofs. He also manages to describe them as formal calculi and to establish their expected meta-theoretical properties. In addition to the

theoretical contributions of this thesis, the author developed a large code-base and successfully implemented a realistic and usable graphical interface for ITPs (**Actema**) as well as a plugin to connect it to the Coq proof assistant (**coq-actema**). This is a very nice piece of software engineering, which combines advanced automated and/or interactive theorem proving technologies with web technologies.

In conclusion, I want to underline that the work achieved in this thesis is of paramount importance to make interactive theorem provers such as Coq easier and faster to adopt and use efficiently by newcomers, regardless of their initial level of expertise. Therefore I strongly support that the author of this very nice work be awarded *le grade de Docteur en informatique de l'Ecole Polytechnique*.

Strasbourg, April 24th 2024

Nicolas MAGAUD

A handwritten signature in black ink, reading "N. Magaud". The signature is written in a cursive style with a long horizontal stroke at the end.